

Database Schema / Data Definition Document (DDD)

Introduction

A **Database Schema / Data Definition Document (DDD)** provides a detailed blueprint of the database structure, describing how data is stored, connected, and managed.

It is essential for developers, database administrators, and analysts to understand the relationships between tables and ensure the system is built for scalability, accuracy, and compliance.

In Business Central and Azure SQL environments, the DDD serves to:

- Define **tables and their relationships** for core business processes like Sales, Purchasing, and Inventory.
- Provide a reference for **system integrations** with Power Platform, Azure services, and external APIs.
- Ensure **data integrity** by documenting dependencies and constraints.
- Act as the foundation for **maintenance, migrations, and performance tuning**.

Diagram Components

The DDD diagram focuses on **table-level structures and their relationships**, showing how transactional and master tables connect.

Component	Description
Tables	Entities representing collections of related data (e.g., <code>SalesHeader</code> , <code>SalesLine</code>).
Columns (Fields)	Attributes within each table that store specific pieces of data.
Primary Keys (PK)	Fields that uniquely identify each record in a table.
Foreign Keys (FK)	Fields linking a record to related data in another table.
Relationships	Lines that visually represent foreign key connections between tables.

Schema Groupings	Logical grouping of tables, such as Sales, Inventory, or Reference tables.
-------------------------	--

Modeling Symbol Definitions

Symbol	Representation
Rectangle (Table)	Represents a database table with its fields listed.
Bold Field Name	Indicates Primary Key in that table.
Arrow Line →	Shows a one-to-many relationship between two tables.
Blue Outlined Box	Denotes core transactional tables.
Dashed Line	Represents optional or conditional relationships.

Diagram Example

Below is a **Database Schema diagram** representing the core sales process in Business Central:

- **SalesHeader** connects to **SalesLine** (one-to-many).
- **SalesHeader** links to **Customer** (each order belongs to one customer).
- **SalesLine** connects to **Item** (each line is for a specific item).
- **SalesLine** links to **WarehouseShipmentHeader** for fulfillment tracking.

Data Element Metadata

Data Element Metadata identifies *data* information about individual data elements within a database. Metadata includes attributes such as element's name, data type, validation rules, units of measure, and descriptions. Implementing metadata at the data element level enhances data quality, facilitates data management, and aids in standardizing how data is interpreted and used.

Diagram Components

Data Element	Represents a single, field or attribute within a database table
Metadata	Contains additional information about the data element incl, description, validation, units.
Stored Within	Identifies the table name within which the data element is stored
Table	Represents a database table containing the data element

Diagram Symbol Definitions

Data Element	A rectangular box represents a field or attribute
Data Type	A specification of the kind of data associated with the data element
Description	Brief textual explanation of data element's meaning or purpose
Units	Units, measurement if applicable to the data element
Table	A larger rectangular box as table

Diagram Example

